

CPSC 554K

Design Review Report #1

Immersive VR Training

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1. Introduction

The prevalence of mental health disorders is on the rise worldwide, affecting almost half of the population in middle- and high-income countries and impacting their daily lives in various ways (Trautmann, 2016). To address this issue, many studies have suggested the use of immersive extended reality (XR) technologies as a complement or replacement for traditional pen-and-paper paradigms, to aid in the assessment and treatment of mental health disorders across domains. Immersive technologies like virtual reality (VR) enable clinicians to deliver and control stimuli while capturing responses with high-fidelity assessment tests, resulting in meaningful clinical care that aligns with real-world deficits in cognitive functioning. This approach provides clinicians with acceptable predictions about patients' conditions in real life. Additionally, cognitive rehabilitation and remediation, which involves applying configurable exercises that pair standardized assessment with adaptive and gamified variants for therapy, show potential to improve specific cognitive domains and translate these improvements to real-world functioning through repeated practice.

The Cognitive Health Technologies (CHT) team at the National Research Council Canada (NRC) has developed bWell, a cognitive care platform that provides multisensory cognitive tasks for adoption by treatment providers. Intending to provide immersive 3D environments for cognitive assessment and treatment, bWell serves as a toolkit that comprises eight exercises targeting different cognitive domains. The CHT team first tested and validated the prototype with healthy adults in a laboratory environment, establishing a cognitive care network to provide iterative input throughout the development process (Gagnon Shagetz, 2021). Building on this work, the team then focused on major depressive disorder (MDD) and engaged in a co-design process with patient end-users and clinicians to create a VR cognitive remediation program called bWell-D (Hernandez Hernandez, 2022). Moving forward, the CHT team is now addressing the challenges of determining the effective components and clinical relevance of digital interventions, and the low uptake and high non-compliance rates of digital intervention delivery, through the co-design of content and theory-based gamification.

Through this collaboration with DFP@UBC, the CHT team would like our team to explore the potential of patients using bWell at home for cognitive rehabilitation and remediation. To achieve this objective, we need to understand the general user demands for self-administrative mental health applications and their habits with cognitive training and VR technologies. In addition, as bWell is originally designed and developed for clinical care, the usability of current bWell interfaces has to be evaluated and adjusted to meet the requirements of patients who have little or no previous experience with VR technologies. At last, to ensure users' continuous usage of bWell at home, designs for enhancing long-term engagement also remain to be explored and integrated into current designs.

2. Literature review:

Upon preliminary study, we set two main objectives for this project and identified three main categories to continue our studies. Our objectives are to identify and improve usability issues as well as come up with approaches to maintain long-term engagement. In what follows, we will go over the three areas we explored in our literature review.

2.1. Cognitive behavioral therapy and VR at home:

Miller et al. conducted a literature review on 14 studies published between 2000 to 2012, to examine the feasibility and effectiveness of VR/gaming systems to be used at home. The training necessary to set up and use the required equipment was conducted with the help of user manuals and during supervised orientation sessions at the start of the studies. No study examined post-study use of this technology, and very few identified safety concerns and the cost of equipment. The literature review mentioned that the examined studies provided insufficient information and that most of them described the feasibility of their intervention poorly. The use of VR/gaming activities to improve the cognitive and emotional well-being of participants was rarely investigated; their loneliness, level of social support, or quality of life was addressed only once, and their mood and mental health were never evaluated.

This literature review demonstrated the knowledge gap in evaluating the effect of VR at home. Furthermore, it acknowledged the importance of evaluating feasibility variables, including recruitment, retention, training, assistance, safety, cost, acceptability, and engagement. Considering and evaluating these variables during our project will lead to better results regarding the application of Be-well in the home setting.

2.2. Usability of VR products

Since VR development practices have a diverse set of practices compared to traditional software development, the evaluation processes including usability evaluation are also different and executed unconventionally by industrial practitioners today. Our study relied heavily on a recent systematic literature review of the usability evaluation methods practiced by industrial researchers while building VR products after 2000 (Karre, 2020). According to the study, controlled experiments are the most widely used usability evaluation approach, while other focused approaches like Pluralistic Walk-through, Cognitive Walkthrough, and Subjective or Objective Evaluation, are also primarily implemented for particular sets of end users. Group studies and survey-based empirical approaches are found to help capture the participant feedback to understand the impact of their usability evaluation. Besides, metric-wisely, unique metrics such as Interaction Capacity and Efficacy are suggested for consideration in addition to regular usability metrics.

2.3. Long-term engagement:

Eyal (2014) and Fogg (2009) discuss persuasive design principles that aim to influence user behavior and increase engagement with a product. Eyal argues that companies can form user habits by guiding them through a sequence of experiences called "Hooks." The Hook Model consists of four steps: triggers, actions, variable rewards, and investment. Triggers can be internal or external, and the action should be simple and easy to complete. The reward should be fulfilling and leave the user wanting more. Finally, investment requires the user to put in some effort, which increases the likelihood of a return. On the other hand, Fogg's Behavioral Model (FBM) focuses on the combination of motivation, ability, and triggers. Behavior (B) happens when these three elements come together at the same moment. Motivation refers to the drive that prompts individuals to modify their behavior, which can be classified into three levels: Sensation, Anticipation, and Belonging. Ability refers to the ease of performing a task at a given moment, and it can be increased by increasing simplicity. Triggers are the prompts that initiate user action. By utilizing these principles, companies can increase user engagement and create habit-forming products. The use of email or phone notification for a reminder can act as an external trigger or providing a user's progress via email or an app is a low-cost way to keep the users motivated; thus, increasing long-term engagement.

3. Hypothesis / Research Questions

- What are the usability problems people have for B-well at home (hardware and software)?
- How do we maintain and encourage long-term engagement?

4. Method

4.1 Overview:

Cognitive walkthrough is a usability evaluation methodology that assesses the learnability of interfaces. This method evaluates whether a user can complete a task for the first time and, if so, whether they do so successfully and efficiently. The cognitive walkthrough methodology emphasizes the importance of design elements that assist a user's ability to learn the interface, such as providing feedback and using common interaction patterns. This approach involves a detailed analysis of the user's actions, assumptions, and decision-making processes throughout the task, which was captured using the think-aloud method and a follow-up semi-structured interview. By conducting a cognitive walkthrough, we could gain insight into how new users interact with the interface, what they expect from it, what difficulties and challenges they face, and how they learn it.

4.2 Study Materials:

We conducted a semi-structured interview lasting 25-30 minutes. The interviews were conducted in English.

4.3 Procedure:

Low-Fidelity Prototype was developed by using screenshots from a recorded orientation meeting. We aimed to create a Figma prototype to conduct our study since we were yet to have access to a headset to test. Changes in perspective and low resolution were inevitable results of this approach. As an attempt to map the controller input on a 2D screen, an additional B button was added on the bottom right corner of each page.

4.4 Data storage and management:

For this study, we used the UBC-secured zoom software with a custom license for UBC. “It complies with the BC Freedom of Information and Protection of Privacy Act (FIPPA) because the survey data is kept secure and is stored and backed up in Canada.” ([UBC, 2017](#)). All data was stored locally and distributed between the team via UBC emailing and one drive.

4.5 Study Protocol, Tasks, and interview:

A user study protocol was developed, which acted as a guide and contained a layout of the study, introduction, tasks, and interview questions. For more efficient testing, the flow was broken into 3 steps, including onboarding, tutorial, and exercise, each with its own task examples and scenarios. The study was mostly audio/video recorded and was conducted through UBC Zoom, which lasted about 30-40 minutes. Participants were asked to complete each task while thinking aloud. Following the observation, each was asked a set of questions on demographic information, a Likert-scale survey for usability, and general mental health and long-term engagement. The interview was a semi-structured one; thus, the team had the privilege of asking participants to expand on potential interesting points from the observation or the interview itself.

4.6 Participants and Recruitment:

A total of 7 participants were recruited with a male-to-female ratio of 5:2. To quickly gather data, we conveniently sampled from within our network—reaching out to participants using mostly based in Canada via Slack, Discord, UBC Graduate Email listing, and word-of-mouth. Our network included professionals (school and work) and personal (friends and family) connections. Our inclusion criteria were students or past students ages 18 years or older who are English literature of any educational background. Before conducting the interview, informed consent was taken, and all the participants were made aware of keeping their privacy and confidentiality.

5. Results

An inductive qualitative content analysis (Appendix F) was performed on the interview data and open-ended questionnaire questions. For the content analysis, the student researchers first divided texts into consolidated units of words or sentences based on their meaning. Each unit was then labeled with a code by each student researcher individually. These codes were then grouped together by the student researchers collaboratively based on the underlying meaning and relationship with other codes. Questionnaire data were analyzed using IBM SPSS Statistic software. Categorical variables were described as frequency rates and percentages, and continuous variables were described using mean and standard deviation (SD) or median and range. All authors and members of the research team are graduate students with some experience in usability testing. Two of the members of the research have some experience with Virtual reality, and the other two have experience with cognitive therapy, allowing for varied perspectives and ideals. This experience and the unique position of the authors may contribute to the selection and proposition of the codes and themes.

5.1. Demographic Characteristics:

Seven individuals, with the age range of 25-42, participated in this study. Most participants pay extra effort into learning new technologies. Five participants have used VR once in their life. Of the other two, one has a research project on VR, and one has just heard about it. None of the participants owned a VR headset. A summary of participant demographic data and their technology readiness variables can be found in Appendix G.

5.2. Cognitive walkthrough:

The cognitive walkthrough has 3 main steps, including onboarding, tutorial, and exercises. The extracted themes of each step were listed in Appendix F with participants' quotes. Also, Appendix H provides a visual demonstration of the Participants' comments.

5.2.1. Onboarding:

During the cognitive walkthrough of the onboarding task, 2 main themes were extracted based on participants' feedback: a) *The need for more information* b) *Do we really need them?* In the following paragraph, we will go through these themes.

In this onboarding task, individuals have to select their preferences in the settings. Participants expressed that they are not sure which criteria they need to base their selection on. For example, one participant (P6) was confused about the 'eye level cm': "*that's [the eye level cm] not something I know by heart; I've never heard of that. Can I just take a guess?*" Another participant (P4) asked about the avatar and didn't know if the chosen setting should resemble their actual appearance in the real world or could also be based on how they would like to look in the game. Additionally, three participants reflected after playing the game that they never saw the avatar or their chosen skin color during the games, bringing up the question of the relevancy of their choice in the onboarding process: "*What was the avatar selection for? I didn't see myself a single time during the game*" (P7). Thus, options that need to be explained more are 1) eye level cm, 2) body type, and 3) skin color.

5.2.2. Tutorial:

Two main themes emerged during the cognitive walk-through of this phase, 1) *Page Navigation*, and 2) *The Difficulty of Reading*

At the beginning of the study, participants were instructed about the location of the 'B' button, however, during the tutorial, 3 participants didn't know how or when to use that button to navigate different pages, for example, at the end of the tutorial they were unsure of what to do and didn't know they had to press 'B' to go back to the menu. One participant (P1), after finishing the tutorial, mentioned: "Ok, I got the concept of the game, but what do I do now? How do I get back to playing this game?" Also, 2 participants were confused about what tutorial they are supposed to go through because in our current prototype, we, on the one hand, have an actual tutorial which players can choose in the main menu, and, on the other hand, all games have the option to be turned into a tutorial by moving a slider on their respective settings page: "*Why are we going to the theater game? There was a tutorial in the top left of the game selection menu. Would that tutorial have been any different?*" (P3). Another participant complained about the amount of text on the screen during the tutorial: "That's a lot to read, am I supposed to memorize all of that? What happens if I forget something?" (P6) Additionally, they remarked on the low resolution of the text during the tutorial; however, this can be attributed to the quality of screenshots of our low-fidelity prototype and is not expected to be an issue in the finished product.

5.2.3. Exercise step:

In this step, we identified three main themes, including 1) *The Need for More Information*, 2) *A Reminder of Tutorial Information*, and 3) *The Need For Feedback*.

Similar to the situation during the onboarding, participants again expressed the need for more information during the exercise. Mainly, they were curious about the meaning of the option 'table offset': "*What's the purpose of that option? Which offset are we talking about? What does that mean?*" (P3). Furthermore, the same participant (P6) who commented on the length of the text during the tutorial again mentioned the necessity of a reminder of the mechanics of the game in case players couldn't remember every single detail: "*Actually, I don't*

fully remember everything I read earlier on. Is there an option for me to go back or just display it mid-game real quick?”. Last, a point brought up by several participants was the missing feedback. One participant was curious about the points necessary to reach the next level, while another asked for audio/visual feedback in case of successful or failed attempts at getting the right answer: “Is there any confirmation whenever I get something right? Maybe a short sound clip? How do I know if I beat a level; is there gonna be some sort of victory screen?”.

5.3. Usability feedback:

Participants generally gave positive feedback toward the interface design. Appendix I shows the mean score of each item of the questionnaire. Most of the participants indicated that they agree with positive statements on learnability, enjoyability, etc., resulting in a mean score of more than 4 (Agree), while the mean score for the negative statement on interface complexity was less than 3 (Neutral).

5.4. Motivations behind long-term engagement:

Only 3 out of 7 participants already used online self-help tools for mental health, such as motivational youtube channels and mobile cognitive applications. These tools help them to be more calm, motivated and concentrated on their daily tasks, which entices our participants to engage in their favorite tools on a long-term basis.

In the last part of our interview, we asked our participants about the application they most frequently use in their free time. Most of them mentioned social media, with a mean time of 8 hours per week. The three following themes were extracted from the interviews about the motivation for using these apps.

- *Seeing the effect or progress:* Participants mentioned that if they experience a positive outcome or progress toward their aim, they have the motivation to use the tools more.
- *Used long enough to form a habit:* Some others indicated that the long-term usage of an application leads to a habit of using it. As one of the participants mentioned, “[...] habit. I sometimes find myself opening the app without thinking” (P7).
- *A moderate amount of reminder notifications and emails as triggers:* When we asked our participants about the role of reminders in their motivation to use a specific application, they mentioned that it depends on the type of notification, frequency, and content of the notification. They generally preferred Email notifications compared to other types of notifications as they can check them on a daily basis. However, when notifications occur more frequently than 1 to 3 times a week, as mentioned by several participants, which makes them resemble spam, they prefer to mute the notification. One of the participants mentioned, “[It] Probably would be a great reminder to use. I also check my email a couple of times a day. As long as it is not spamming me, I would keep them” (P7). The other factor mentioned by the participants was the content of the notification. If they give them some helpful information about their progress and help them to plan forward or reach a goal, they prefer to keep the notification on.

6. Limitation, Conclusion, and Next Steps:

Not having access to a VR headset for conducting the study seriously limited the outcome of this study as the 2D representation of the app had limited interaction, confusing change of perspective between screens, and the controller mapping button proved to be challenging. Further, due to ethical limitations, we were unable to recruit participants from the targeted user group which left us with limited insight into the mental health needs and preferences of this group.

Despite limitations, this study helped us to evaluate the usability interface of the project and identify the interface parts that need more development. Generally, participants provided positive feedback on the interface design, and this study was a preliminary look into the usability issues. Moving forward, we will focus on *Usability Heuristics evaluation* and a *Cognitive Walkthrough* with the VR headset which will enable us to test the product with the intended medium. We will recruit both expert and naive users for a convenient breadth of data, both from the software point of view and hardware.

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Karre, S. A., Mathur, N., & Reddy, Y. R. (2020). Understanding usability evaluation setup for VR products in industry: A review study. *ACM SIGAPP Applied Computing Review*, 19(4), 17-27.

Appendix A:

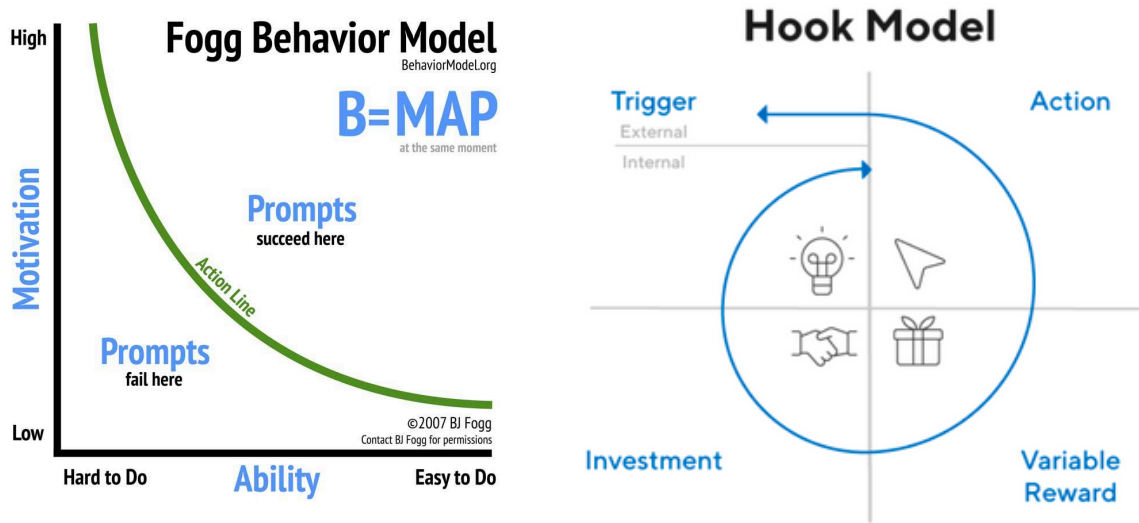


Figure 1 - [From left to right] Fogg model and hook model

Appendix B: Figma prototype



Figure 2 - Figma prototype

Appendix C: Sample screen from the prototype



Figure 3 - Sample screen from the prototype and the placement of “B” button

Appendix D: Consent Form

Link to the consent form: [link](#)

Appendix E: Questionnaire

Link to the questionnaire and protocol: [link](#)

Appendix F: Content Analysis

Categories	Themes	Selected Quotes
Onboarding Step	<i>The Need for More Information</i>	“that’s [the eye level cm] not something I know by heart; I’ve never heard of that. Can I just take a guess?” (P6)
		“The Avatar and skin color should resemble my actual appearance in the real world?” (P4)
	<i>Do We Really Need Them?</i>	What was the avatar selection for? I didn’t see myself a single time during the game”(P7)
Tutorial Step	<i>Page Navigation</i>	“Ok, I got the concept of the game, but what do I do now? How do I get back to playing this game? [from the end of tutorial page]”(P1)
		“Why are we going to the theater game? There was a tutorial in the top left of the game selection menu. Would that tutorial have been any different?” (P3).
	<i>The Difficulty of Reading</i>	“That’s a lot to read, am I supposed to memorize all of that? What happens if I forget something?” (P6)
Exercise Step	<i>The Need for More Information</i>	“What’s the purpose of that option [Table top offset cm]? Which offset are we talking about? What does that mean?” (P3)
	<i>A Reminder of Tutorial Information</i>	“Actually, I don’t fully remember everything I read earlier on. Is there an option for me to go back or just display it mid-game real quick?” (P6)
	<i>The Need For Feedback</i>	“Is there any confirmation whenever I get something right? Maybe a short sound clip? How do I know if I beat a level; is there gonna be some sort of victory screen?” (P2)
Motivations behind long-term engagement	<i>Seeing the effect or progress</i>	“I think if it is tied to assessment, I may use more, if just normal notification, may not be that helpful. It should tell you well.” (P4)
	<i>Used long enough to form a habit:</i>	“[...] habit. I sometimes find myself opening the app without thinking” (P7)

*A moderate amount
of reminder
notifications and
emails as triggers*

“[I]
Usually turn them off as I don’t want to be
distracted during the day. But if
it was something I cared about, I would
leave it on. [It] Probably would be
a great reminder to use. Notifications are
popular. I also check my email a couple of
times a day. As long as it is not spamming
me, I would keep them“ (P7)

Appendix G: Demographic Characteristics

Frequency Percent (%)

Age

Mean: 29.4(±3.4)

Median: 27

Range: 7

Min: 25

Max: 42

Gender

Male	6	86
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Female	1	14
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Employment status

Graduate Student	6	86
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Employee	1	14
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Openness to learning a new technology

Definitely won’t learn	0	0
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Try to learn if it’s essential to life	0	0
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Learn it with medium effort	0	0
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Pay extra effort to learn new technologies	5	71
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	Interested in new technology and follow the trend actively	2	29
Previous experience with learning a new technique	Quite difficult and frustrating	0	0
	Difficult	0	0
	Neutral	2	28
	Easy	3	43
Familiarity with VR technologies	Quite easy and inspiring	1	14
	Never heard about it	0	0
	Have heard about it	0	0
	Have used it a number of times	5	71
	Use it frequently	2	28
	Expert (serious users/was or is in development)	0	0
Have you used a VR headset	Yes	4	57
	No	3	42

Appendix H: Visual demonstration of Participants' comment

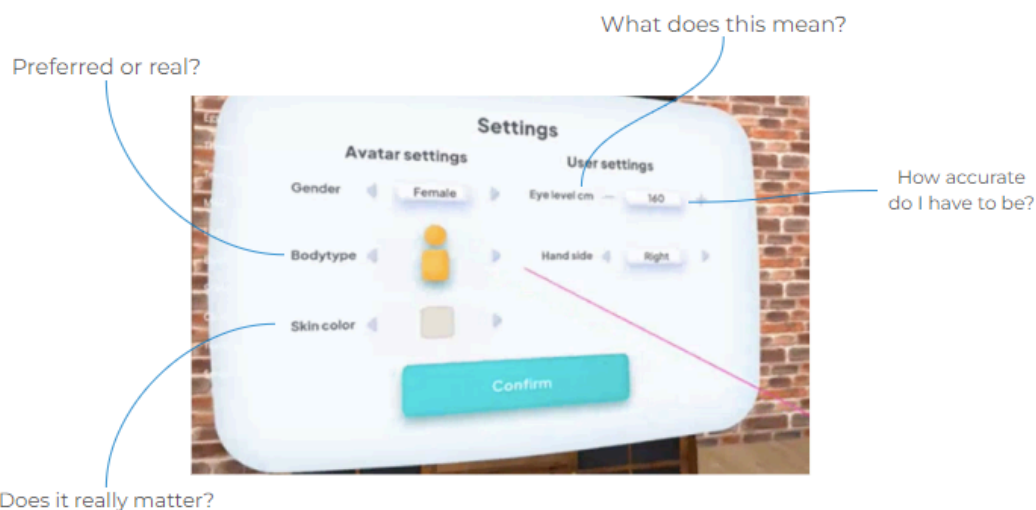


Fig4: Selected Usability feedback- Onboarding

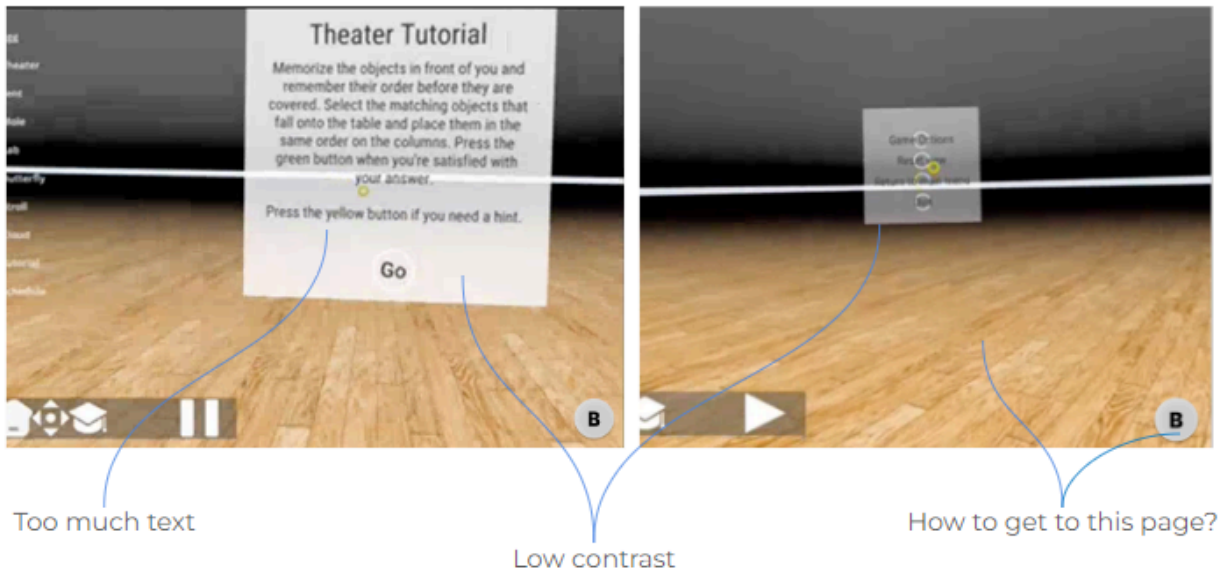


Fig5: Selected Usability feedback- Tutorial

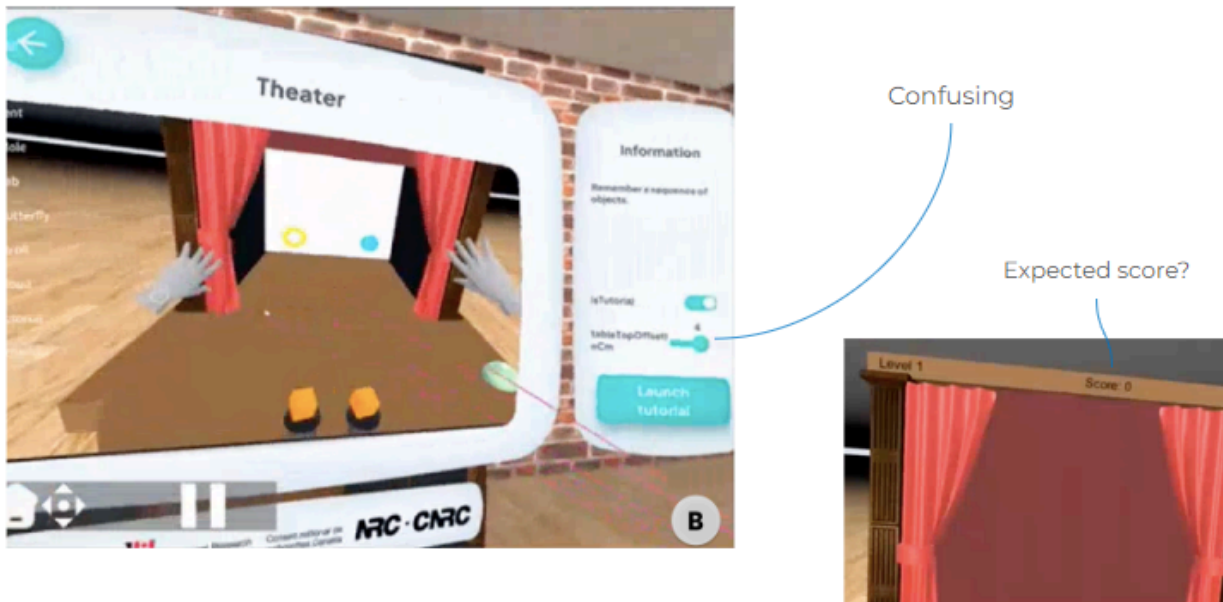


Fig6: Selected Usability feedback- Exercise

Appendix I: Mean score of each item of the questionnaire-Usability study	
	Mean (SD)
I felt very confident using the interfaces	4.0 (0.6)
I found the prototype enjoyable to use	4.0 (0.8)
I found the interfaces unnecessarily complex	2.6 (1.1)
I found various functions in the interfaces well-integrated	3.2 (0.7)
I imagine that most people would learn to use this function very quickly	4.2 (0.7)
I think I would like to use this function frequently	4.0 (0.6)
I think the necessary information and possible actions were well-placed in the interface	4.2 (0.4)

Appendix J: Team contribution	
All members were active in searching the literature, designing the study, and conducting interviews and transcription.	
Team member	Role
Pegah	• Wrote part of the literature review • Analyzed the qualitative and quantitative data • Wrote the result section • Wrote the conclusion part • Wrote part of the next step • Designed tables in the appendix • Formatted the report structure
Himani	• Designed the questionnaire • Wrote the method part • Helped in Collecting Literature review • Editing of report
Merry	• Wrote a part of literature review • Designed study protocol, task scenarios, and interview questions • Made Figma prototype

	• Wrote the method • Wrote and revised limitation, conclusion, and next steps
Zhe	• Developed the consent form • Formatted the report structure and task assignment • Wrote the introduction • Wrote part of the literature review • Wrote part of the next step • Finalized the reference format